Integration from Standard Integrals

Q1. Find an antiderivative of each of the following:

(a) $7x^3 + 6x^2 - 4x + 3$ (b) $\sin(\pi x)$ (c) e^{3x}

(d) $\dfrac{3}{2x^2}$ (e) $\cos\left(\dfrac{2\pi x}{3}\right)$ (f) $\dfrac{4}{x}$

(g) $\sec^2\left(\dfrac{x}{2}\right)$ (h) $2x - \dfrac{1}{2x}$ (i) $e^{-\frac{x}{5}}$

Q2. Equation of Curve. The slope of a certain curve at any point (x, y) on the curve is given by $(x-1)^3 - 2$, and the curve passes through the point $(3, 0)$. Find the equation of the curve.

Q3. Inverse Trigonometric. Integrate the following:

(a) $\dfrac{-2}{\sqrt{16 - x^2}}$ (b) $\dfrac{4}{25 + x^2}$ (c) $\dfrac{7}{\sqrt{3 - x^2}}$

Integration by Substitution

Q4. Find each of the following antiderivatives:

(a) $\displaystyle\int 2x(x^2+1)^5 \, \mathrm{d}x$ (b) $\displaystyle\int 3x^2 \cos(x^3+5) \, \mathrm{d}x$ (c) $\displaystyle\int \frac{x}{(x^2+1)^3} \, \mathrm{d}x$

(d) $\displaystyle\int (8x+6)e^{2x^2+3x} \, \mathrm{d}x$ (e) $\displaystyle\int 5x\sqrt{9+x^2} \, \mathrm{d}x$ (f) $\displaystyle\int \frac{6\sin(3x)}{\cos^2(3x)} \, \mathrm{d}x$

(g) $\displaystyle\int \frac{5}{x} \sin(\log(x)) \, \mathrm{d}x$

Q5. Complete the Square. Find antiderivatives of the following:

(a) $\dfrac{1}{x^2 + 2x + 2}$ (b) $\dfrac{1}{\sqrt{-x^2 - 4x + 21}}$ (c) $\dfrac{-1}{\sqrt{7 - x^2 + 6x}}$

Q6. Linear Substitutions. Evaluate the following:

(a) $\displaystyle\int x\sqrt{1-x}\,\mathrm{d}x$

(b) $\displaystyle\int \frac{2x-1}{(x-1)^2}\,\mathrm{d}x$

(c) $\displaystyle\int (x+3)(x+4)^{\frac{1}{3}}\,\mathrm{d}x$

(d) $\displaystyle\int (2x+1)(x+3)^{20}\,\mathrm{d}x$

Integration by Partial Fractions

Q7. Partial Fractions. Antidifferentiate the following:

(a) $\dfrac{3x}{(x-2)(x+4)}$

(b) $\dfrac{3}{x^2-5x+4}$

(c) $\dfrac{2x+1}{x^2+4x+4}$

(d) $\dfrac{x+3}{x^2-3x+2}$

(e) $\dfrac{2x+1}{x^2-1}$

Q8. Long Division. Find antiderivatives of the following, after first applying polynomial long division.

(a) $\dfrac{x^3+x^2-3x+3}{x+2}$

(b) $\dfrac{4x^3+8x^2+5x+13}{4x^2+5}$

(c) $\dfrac{x^3+3}{x^2-x}$

(d) $\dfrac{x^3+4x^2+11x+2}{x^2+4x+10}$

(e) $\dfrac{4x^3+42x+1}{x^2+10}$

Q9. Find the following:

(a) $\int \cos^2(x)\sin^3(x)\,\mathrm{d}x$ (b) $\int \frac{x^3+3x-2}{x^2-x}\,dx$ (c) $\int \cos(2x)\,e^{\sin(2x)}\,dx$

(d) $\int \cos^4(x)\,dx$ (e) $\int \frac{1}{x^2+6x+10}\,\mathrm{d}x$ (f) $\int \frac{1}{\sqrt{5+x}}\,\mathrm{d}x$

(g) $\int \frac{-2x-3}{x^2-x}\,dx$ (h) $\int \frac{x-2}{\sqrt{x+1}}\,dx$ (i) $\int \frac{1}{8+2x^2}\,dx$

(j) $\frac{2x}{x^2-2x+10}$ (k) $\int \frac{x^4+6x^3+10x^2+11x+13}{x^2+6x+10}\,\mathrm{d}x$

(l*) $\int \tan^2(x)\sec^4(x)\,\mathrm{d}x$ $\left[\text{Hint: Let } u=\tan(x) \text{ and express in terms of } u \text{ and } \frac{du}{dx}.\right]$

Definite Integrals

Q10. Evaluate the following definite integrals:

(a) $\int_0^2 (3x^2+2x+4)\,\mathrm{d}x$ (b) $\int_0^{\pi/4} (\cos(\theta)+2\theta)\,d\theta$

(c) $\int_{-1}^1 (e^t-e^{-t})\,dt$ (d) $\int_1^2 \frac{1}{3t}\,dt$

Q11. Substitution. Evaluate the following:

(a) $\int_0^1 x\sqrt{1-x}\,\mathrm{d}x$ (b) $\int_0^{\pi/4} \sin^3 t\cos t\,dt$

(c) $\int_0^3 x\sqrt{x^2+16}\,\mathrm{d}x$ (d) $\int_e^{e^2} \frac{1}{\log x}\cdot\frac{1}{x}\,\mathrm{d}x$

Q12. Evaluate the following:

(a) $\int_2^3 \frac{3}{x^3}\,\mathrm{d}x$ (b) $\int_0^{\pi/2} \sin^2(2\theta)\,d\theta$ (c) $\int_0^{1/2} \frac{3}{\sqrt{1-x^2}}\,\mathrm{d}x$

(d) $\int_1^2 \frac{1}{x(x+1)}\,\mathrm{d}x$ (e) $\int_2^3 \frac{2x+6}{(x-1)^2}\,\mathrm{d}x$ (f) $\int_{-1}^1 \frac{e^t}{e^t+1}\,dt$

(g) $\int_0^{\pi/4} \sec^2 x\,\mathrm{d}x$ (h) $\int_0^1 \frac{x^3+x^2+4x+1}{x^2+1}\,dx$

(i) $\int_{-\pi/6}^{\pi/3} \frac{\sin x}{10+6\cos x+\cos^2 x}\,dx$

Areas

Q13. Area under the graph. Find the areas indicated below.

(a) Area under $y = x + \dfrac{1}{x^2}$ from $x = 2$ to $x = 3$.

(b) Area under $y = \dfrac{6}{4 + x^2}$ from $x = -2$ to $x = 2$.

(c) Area under $y = \dfrac{2}{x} \log x$ from $x = 1$ to $x = e$.

(d) Area enclosed by $y = -1 + \dfrac{2}{x^2 + 1}$ and the x-axis.

Q14. With respect to x. For each pair of curves below, sketch them and find their intersection points. Find the area enclosed between the curves. (a) $y = x^2$ and $y = 2x$.

(b) $y = 4x^2$ and $y = x^2 + 3$.

(c) $y^2 = x$ and $y = \dfrac{1}{3}x$.

Q15. With respect to y. Sketch the pairs of curves and find the areas enclosed by them.

(a) $x = \sqrt{y}$ and $x = \dfrac{y}{2}$.

(b) $x = y^2$ and $x = 3y$.

(c) $x = -y^2 + 2$ and $y = x$.

Answers

Topic 6: Integration

A1. (a) $\frac{7}{4}x^4 + 2x^3 - 2x^2 + 3x + C$ (b) $-\frac{1}{\pi}\cos(\pi x) + C$ (c) $\frac{1}{3}e^{3x} + C$

(d) $-\frac{3}{2x} + C$ (e) $\frac{3}{2\pi}\sin(\frac{2\pi x}{3}) + C$ (f) $4\log|x| + C$

(g) $2\tan(\frac{x}{2}) + C$ (h) $x^2 - \frac{1}{2}\log|x| + C$ (i) $-5e^{-\frac{x}{5}} + C$

A2. Equation of Curve. $y = \frac{1}{4}(x-1)^4 - 2x + 2$

A3. Inverse Trigonometric. (a) $2\arccos\frac{x}{4} + C$ (b) $\frac{4}{5}\arctan\frac{x}{5} + C$ (c) $7\arcsin\frac{x}{\sqrt{3}} + C$

A4.

(a) $\frac{1}{6}(x^2+1)^6 + C$ (b) $\sin(x^3+5) + C$ (c) $\frac{-1}{4(x^2+1)^2} + C$ (d) $2e^{2x^2+3x} + C$

(e) $\frac{5}{3}(9+x^2)^{3/2} + C$ (f) $\frac{2}{\cos(3x)} + C$ (g) $-5\cos(\log(x)) + C$

A5. Complete the Square. (a) $\arctan(x+1) + C$ (b) $\arcsin\frac{x+2}{5} + C$ (c) $\arccos\frac{x-3}{4} + C$

A6. Linear Substitutions. (a) $-\frac{2}{3}(1-x)^{3/2} + \frac{2}{5}(1-x)^{5/2} + C$ (b) $2\log|x-1| - \frac{1}{x-1} + C$

(c) $\frac{3}{7}(x+4)^{7/3} - \frac{3}{4}(x+4)^{4/3} + C$ (d) $\frac{1}{11}(x+3)^{22} - \frac{5}{21}(x+3)^{21} + C$

A7. Partial Fractions.

(a) $\log|x-2| + 2\log|x+4| + C$ (b) $\log|x-4| - \log|x-1| + C$ (c) $2\log|x+2| + \frac{3}{x+2} + C$

(d) $5\log|x-2| - 4\log|x-1| + C$ (e) $\frac{1}{2}\log|x+1| + \frac{3}{2}\log|x-1| + C$

A8. Long Division. (a) $\frac{1}{3}x^3 - \frac{1}{2}x^2 - x + 5\log|x+2| + C$ (b) $\frac{x^2}{2} + 2x + \frac{3\arctan(2x/\sqrt{5})}{2\sqrt{5}} + C$

(c) $\frac{1}{2}x^2 + x - 3\log|x| + 4\log|x-1| + C$ (d) $\frac{x^2}{2} + \frac{1}{2}\log(10+4x+x^2) + C$

(e) $2x^2 + \log(x^2+10) + \frac{\arctan(x/\sqrt{10})}{\sqrt{10}} + C$

A9. $\left[\text{Hint: Let } u = \tan(x) \text{ and express in terms of } u \text{ and } \frac{du}{dx}.\right]$

(a) $-\frac{1}{3}\cos^3 x + \frac{1}{5}\cos^5 x + C$ (b) $\frac{1}{2}x^2 + x + 2\log|x^2 - x| + C$ (c) $\frac{1}{2}e^{\sin 2x} + C$ (d) $\frac{3x}{8} + \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C$ (e) $\arctan(x+3) + C$ (f) $2\sqrt{5+x} + C$

(g) $3\log|x| - 5\log|x-1| + C$ (h) $\frac{2}{3}(-8+x)\sqrt{1+x} + C$ (i) $\frac{1}{4}\arctan\frac{x}{2} + C$

(j) $\log(x^2 - 2x + 10) + \frac{2}{3}\arctan\left(\frac{x-1}{3}\right) + C$ (k) $\frac{1}{3}x^3 + \frac{11}{2}\log(x^2+6x+10) - 20\arctan(x+3) + C$

(l) $\frac{1}{5}\tan^5(x) + \frac{1}{3}\tan^3(x) + C$

A10. (a) 20 (b) $\frac{1}{\sqrt{2}} + \frac{\pi^2}{16}$ (c) 0 (d) $\frac{1}{3}\log 2$

A11. Substitution. (a) $\frac{4}{15}$ (b) $\frac{1}{16}$ (c) $\frac{61}{3}$ (d) $\log 2$

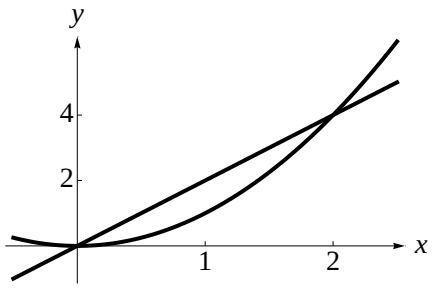
A12. (a) $\frac{5}{24}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\log\frac{4}{3}$ (e) $\log 4 + 4$

(f) 1 (g) 1 (h) $\frac{3}{2}(1+\log 2)$ (i) $-\arctan(7/2) + \arctan(3 + \frac{\sqrt{3}}{2})$

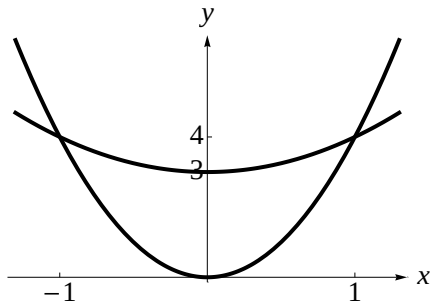
A13. Area under the graph. (a) $\frac{8}{3}$ units^2 (b) $\frac{3\pi}{2}$ units^2 (c) 1 unit^2 (d) $\pi - 2$ units^2

A14. With respect to x.

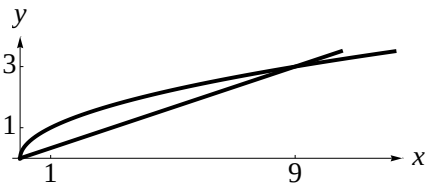
(a) $\frac{4}{3}$



(b) 4

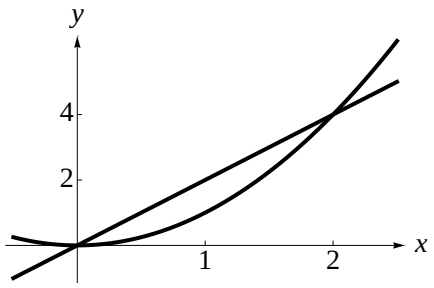


(c) $\frac{9}{2}$

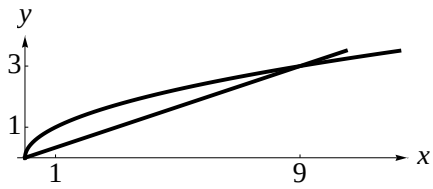


A15. With respect to y.

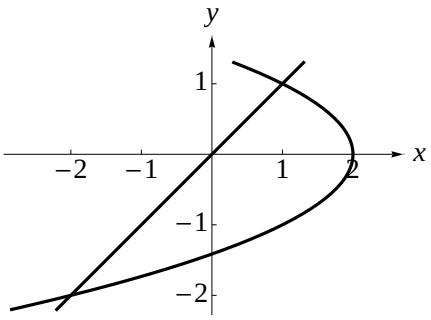
(a) $\frac{4}{3}$



(b) $\frac{9}{2}$



(c) $\frac{9}{2}$



Useful Formulae

Pythagorean identity

$$\cos^2(x) + \sin^2(x) = 1$$

Compound angle formulae

$$\begin{aligned}
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\tan(x+y) &= \frac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}
\end{aligned}$$

Derivatives of inverse trigonometric functions

$$\begin{aligned}
\arcsin'(x) &= \frac{1}{\sqrt{1-x^2}} \\
\arccos'(x) &= -\frac{1}{\sqrt{1-x^2}} \\
\arctan'(x) &= \frac{1}{1+x^2}
\end{aligned}$$

Antiderivatives from inverse trigonometric functions

$$\begin{aligned}
\int \frac{1}{\sqrt{s^2-x^2}}\,dx &= \arcsin\left(\frac{x}{s}\right) + C \\
\int \frac{-1}{\sqrt{s^2-x^2}}\,dx &= \arccos\left(\frac{x}{s}\right) + C \\
\int \frac{1}{s^2+x^2}\,dx &= \frac{1}{s}\arctan\left(\frac{x}{s}\right) + C
\end{aligned}$$

where s is a *positive* constant, and C is an arbitrary constant of integration.

Complex exponential formulae

$$\begin{aligned}
e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\
\cos(\theta) &= \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \\
\sin(\theta) &= \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)
\end{aligned}$$

Vector projections

- The *vector projection* of $\mathbf{v}$ onto $\mathbf{u}$ is $\mathbf{v}_{\parallel} = (\hat{\mathbf{u}} \cdot \mathbf{v})\hat{\mathbf{u}} = k\mathbf{u}$, where $k \in \mathbb{R}$ is the unique solution of $\mathbf{u}\cdot(\mathbf{v}-k\mathbf{u}) = 0$.
- The *vector component of* $\mathbf{v}$ *perpendicular to* $\mathbf{u}$ is $\mathbf{v}_{\perp} = \mathbf{v} - \mathbf{v}_{\parallel}$.

Complex roots

The n-th roots of $w = se^{i\phi}$ are $s^{\frac{1}{n}}\, e^{i(\frac{1}{n}(\phi+2k\pi))}$ for $k = 0, 1, \ldots, n-1$.

Changes in speed

Provided $\mathbf{r}'(t) \neq \mathbf{0}$, the speed function $\|\mathbf{r}'(t)\|$ is decreasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) < 0$ and increasing when $\mathbf{r}'(t)\cdot\mathbf{r}''(t) > 0$.